Aula Conectada

Investment case · Bolivia

Hernando Grueso, PhD
UNICEF

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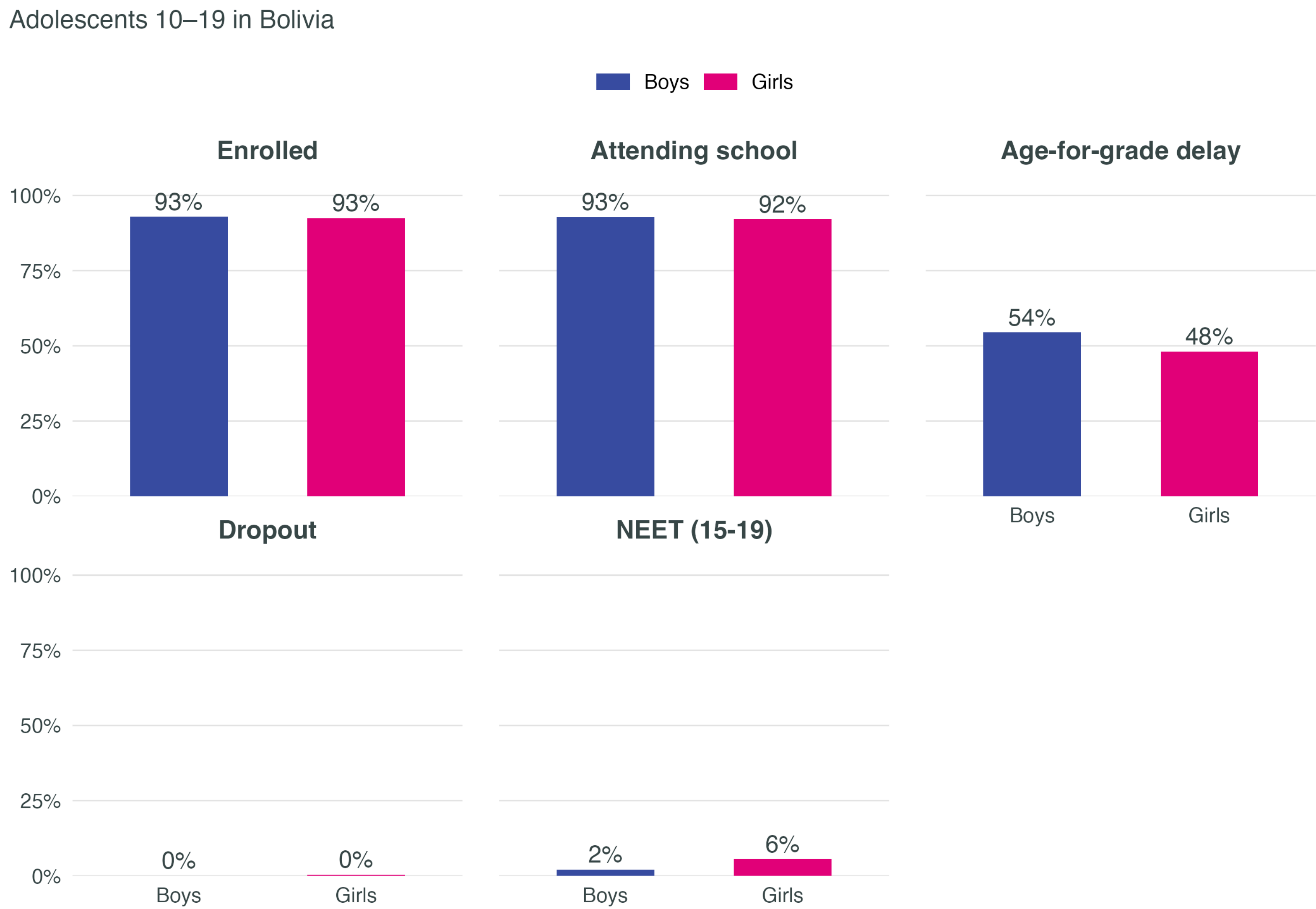
1 · Diagnostic — gaps in education

Target population — adolescent girls 10–19 in Bolivia

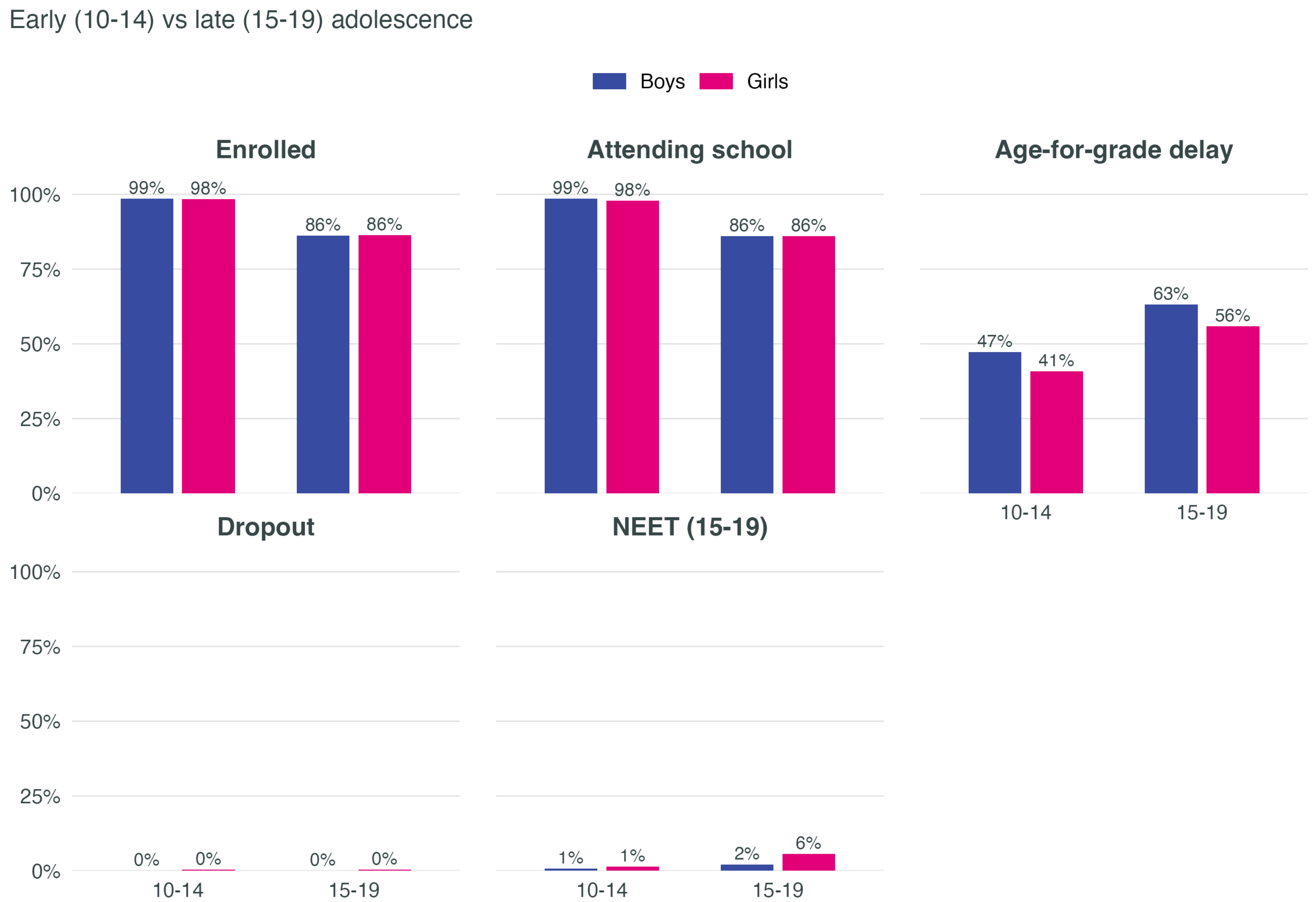
Segment	Weighted N	% of girls 10–19
All adolescents 10–17 (weighted)	1,982,102	203.0%
All girls 10–17 (weighted)	976,569	100.0%
Girls 10–17 — rural	281,797	28.9%
Girls 10–17 — Indigenous	146,455	15.0%
Girls 10–17 — poor (INE)	437,692	44.8%
Girls 10–17 — disability (WG 3+)	4,142	0.4%
Girls 10–17 — no HH internet	215,766	22.1%
Girls 10–17 — priority (rural OR Indigenous OR poor OR disability)	567,836	58.1%

Gender gaps in education



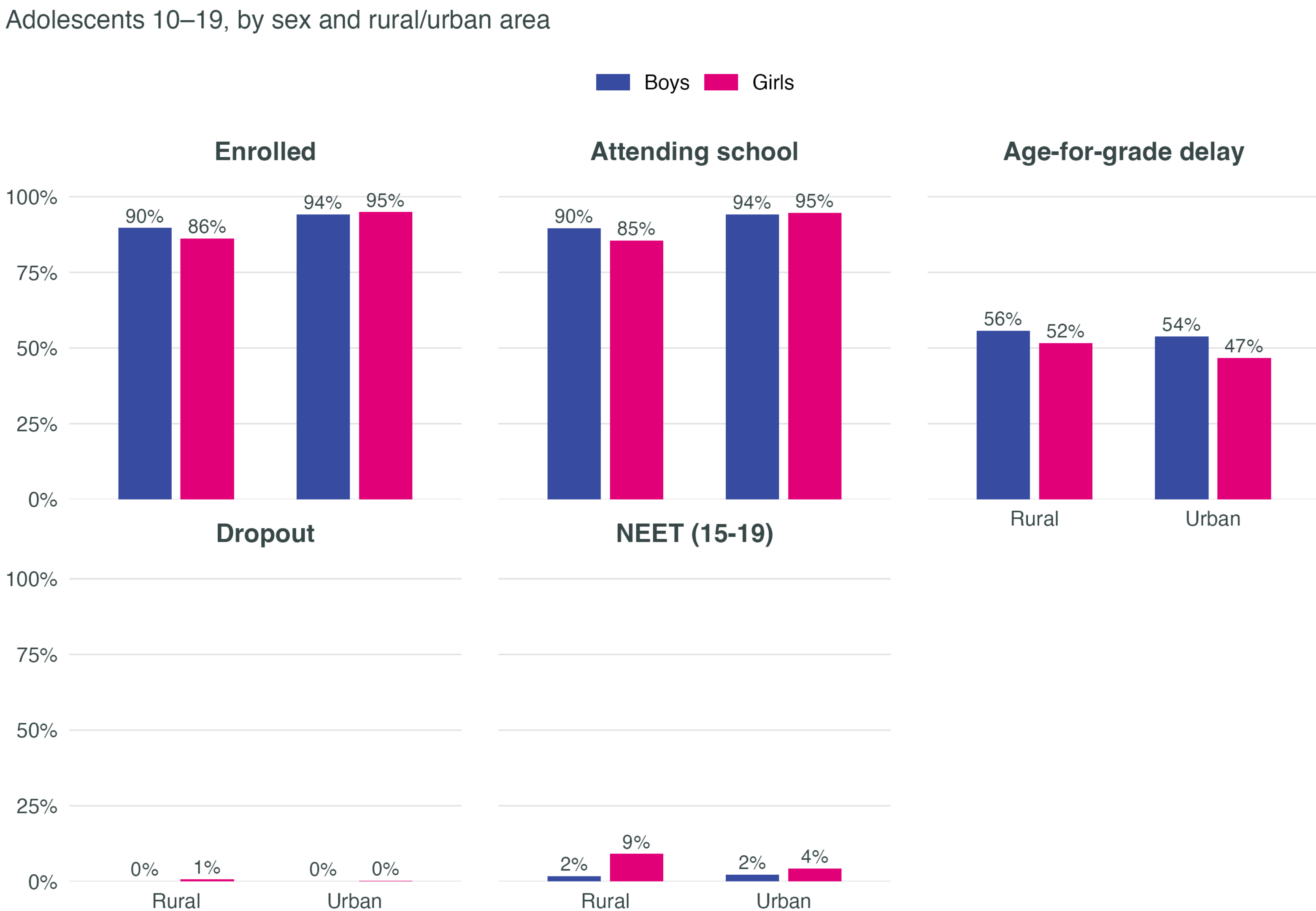
Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19.

Gender gaps by adolescent stage



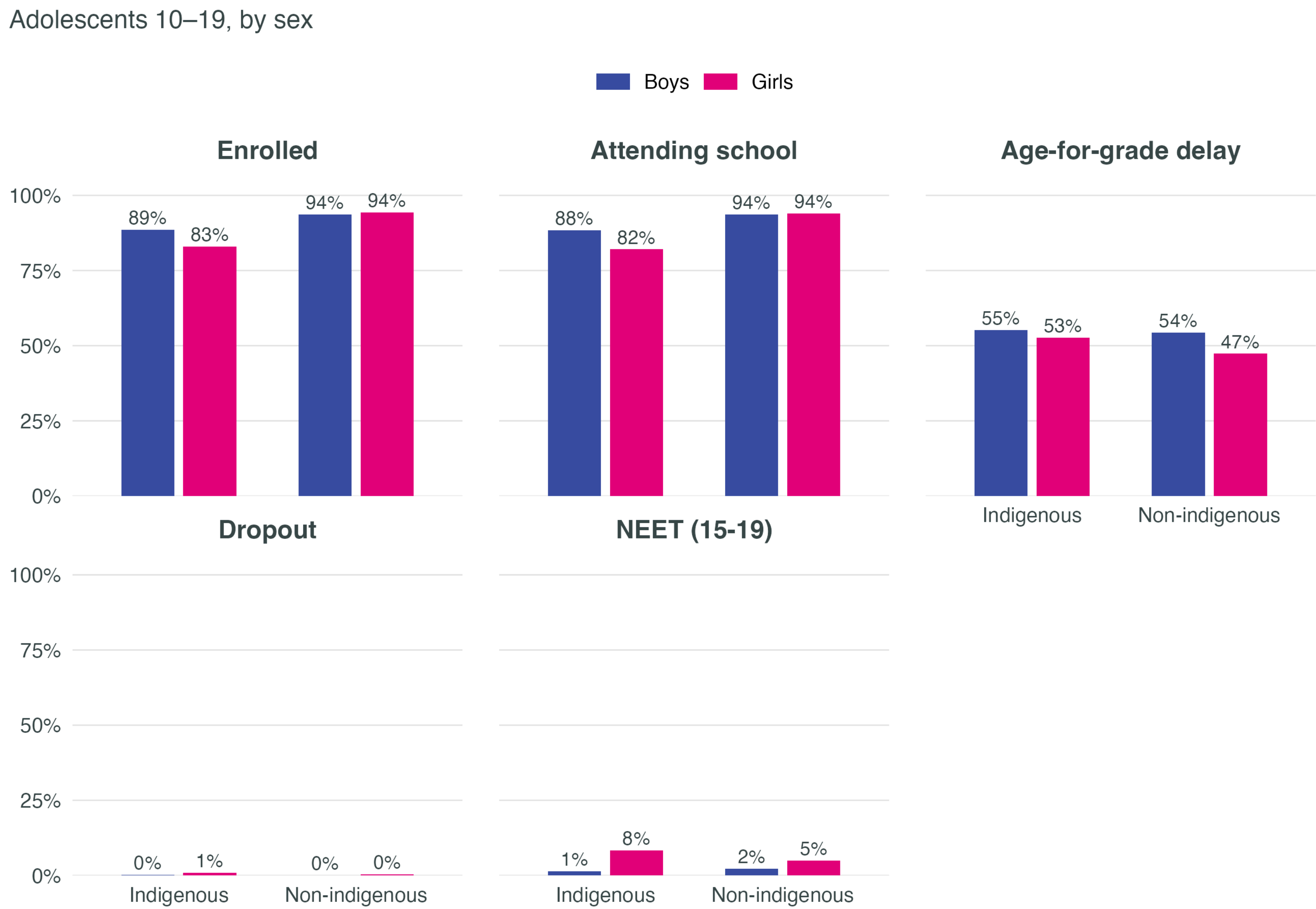
Source: INE Bolivia – Household Survey 2024. Weighted estimates.

Education outcomes by area of residence



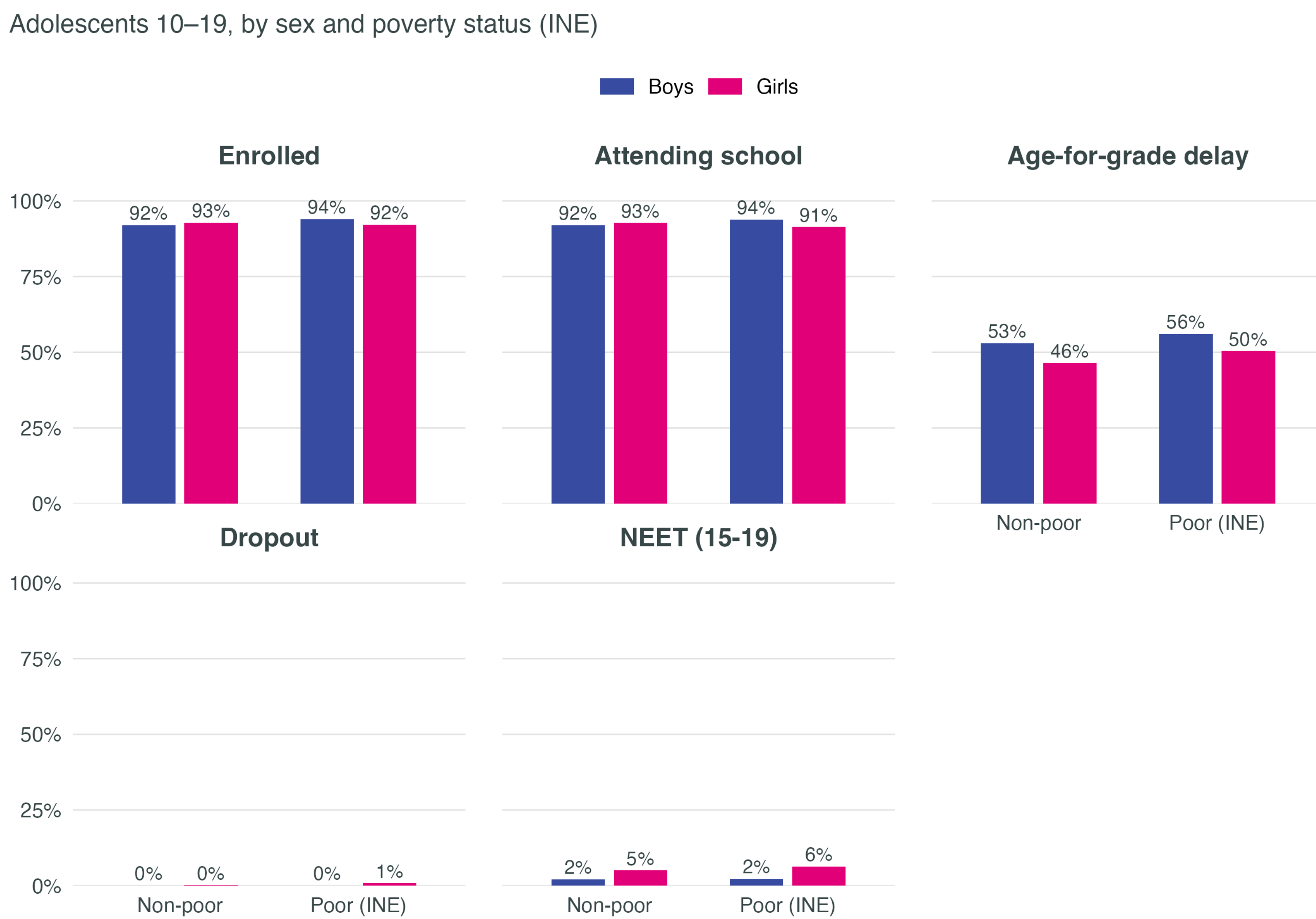
Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19.

Education outcomes by indigenous status



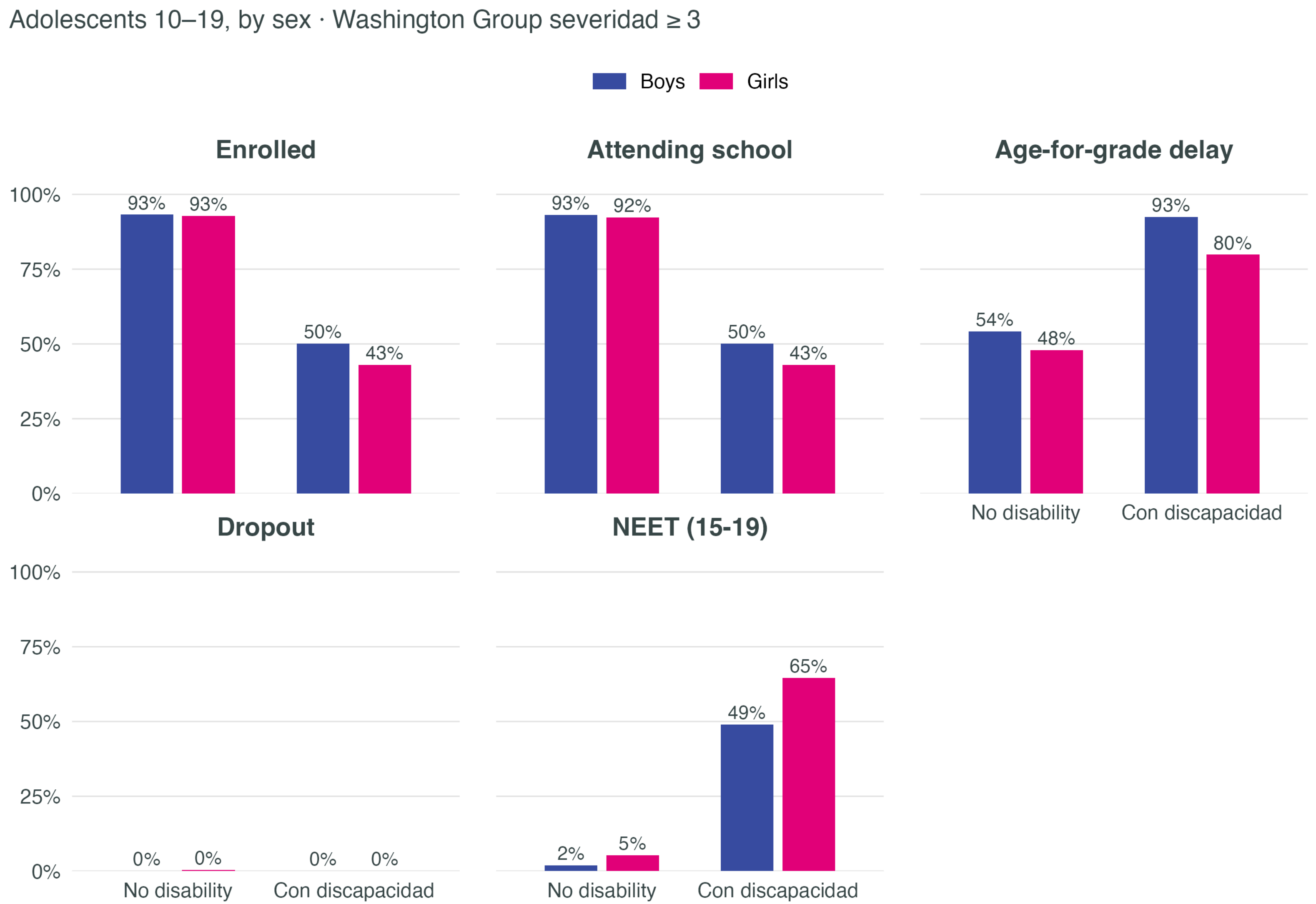
Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19.

Education outcomes by poverty status



Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19.

Education outcomes by disability



Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19. Small sample size (~0.4%) — interpret with caution.

Digital access is a territorial and socioeconomic gap

EH 2024 data are at the household level and therefore do not allow direct measurement of girls' and boys' individual access. At the household level, differences by sex of the adolescent are minimal. **The relevant differences are observed between rural and urban households, indigenous and non-indigenous, and by poverty status.**

Dimensions of the digital divide

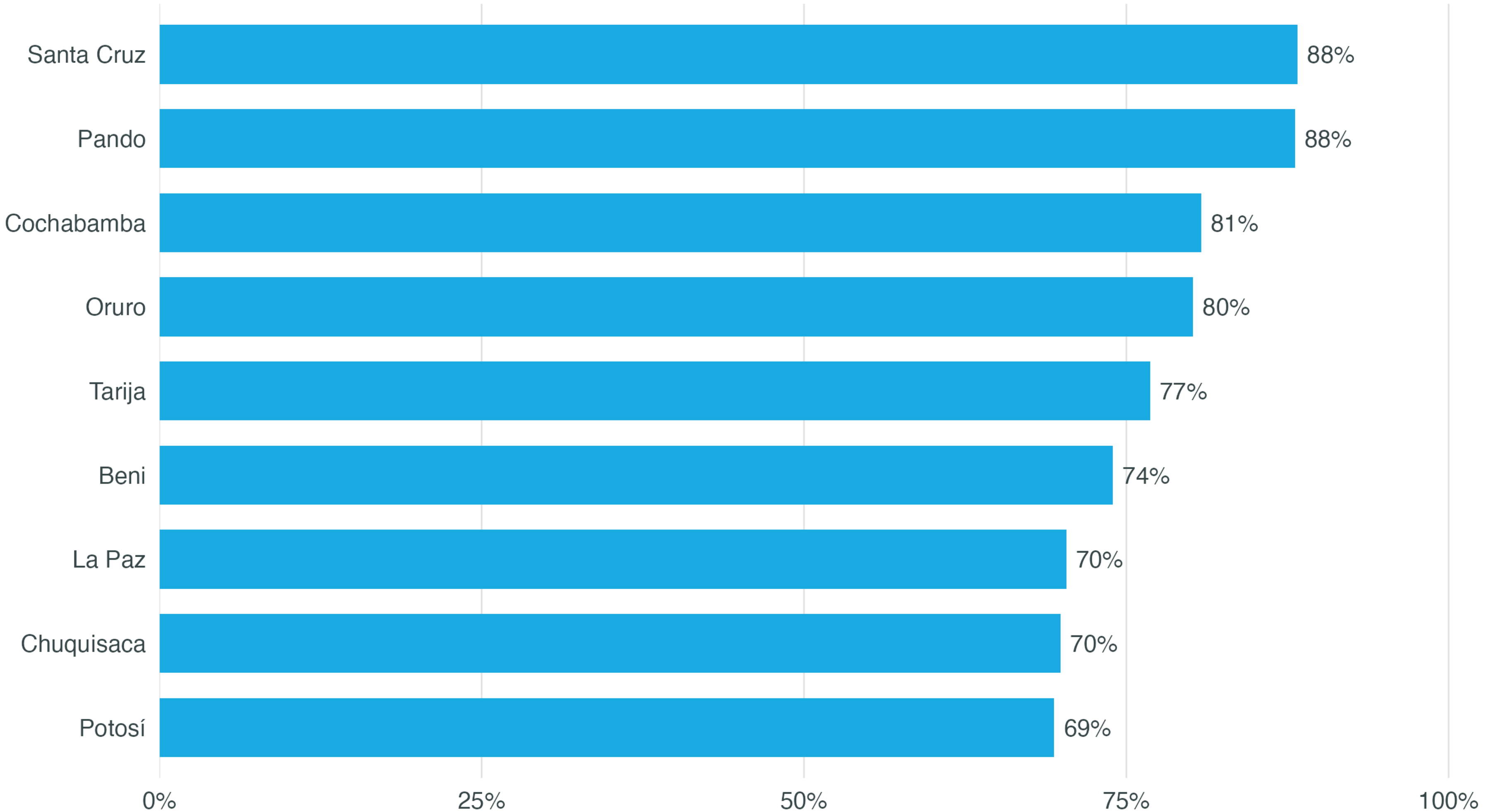
- Connectivity and infrastructure in rural areas
- Device access in indigenous households
- Economic affordability of internet and devices
- Digital skills and meaningful use

Implication for the programme

- Determining whether a gender digital divide exists at the individual level would require data on use, skills, and autonomy that EH 2024 does not capture
- At the household level, access gaps concentrate in the same subgroups that face the largest education gaps
- Aula Conectada acts on institutional access and pedagogical use, not on household access

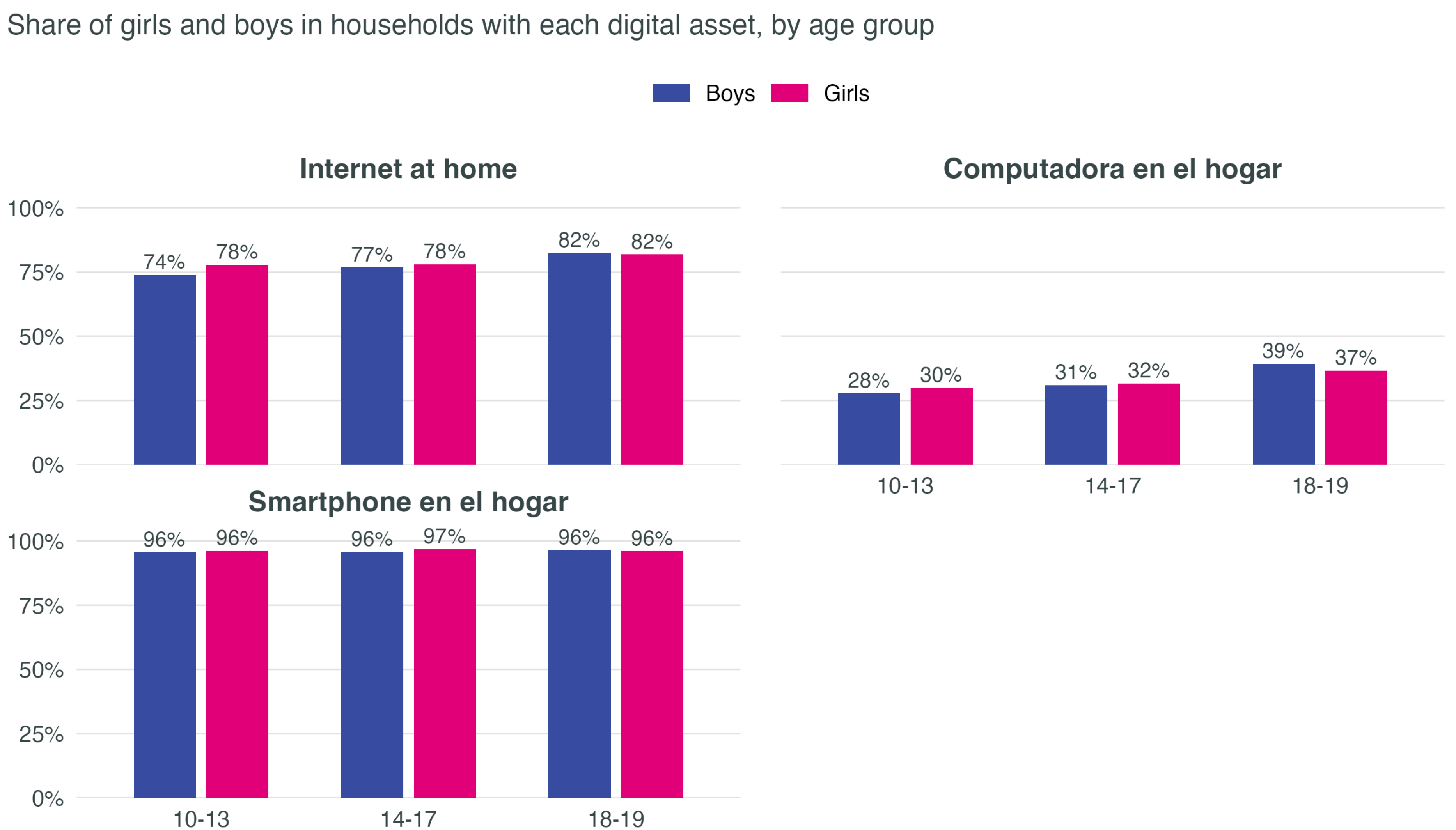
Digital divide — departmental variation

Household internet, adolescent girls 10–19, by department



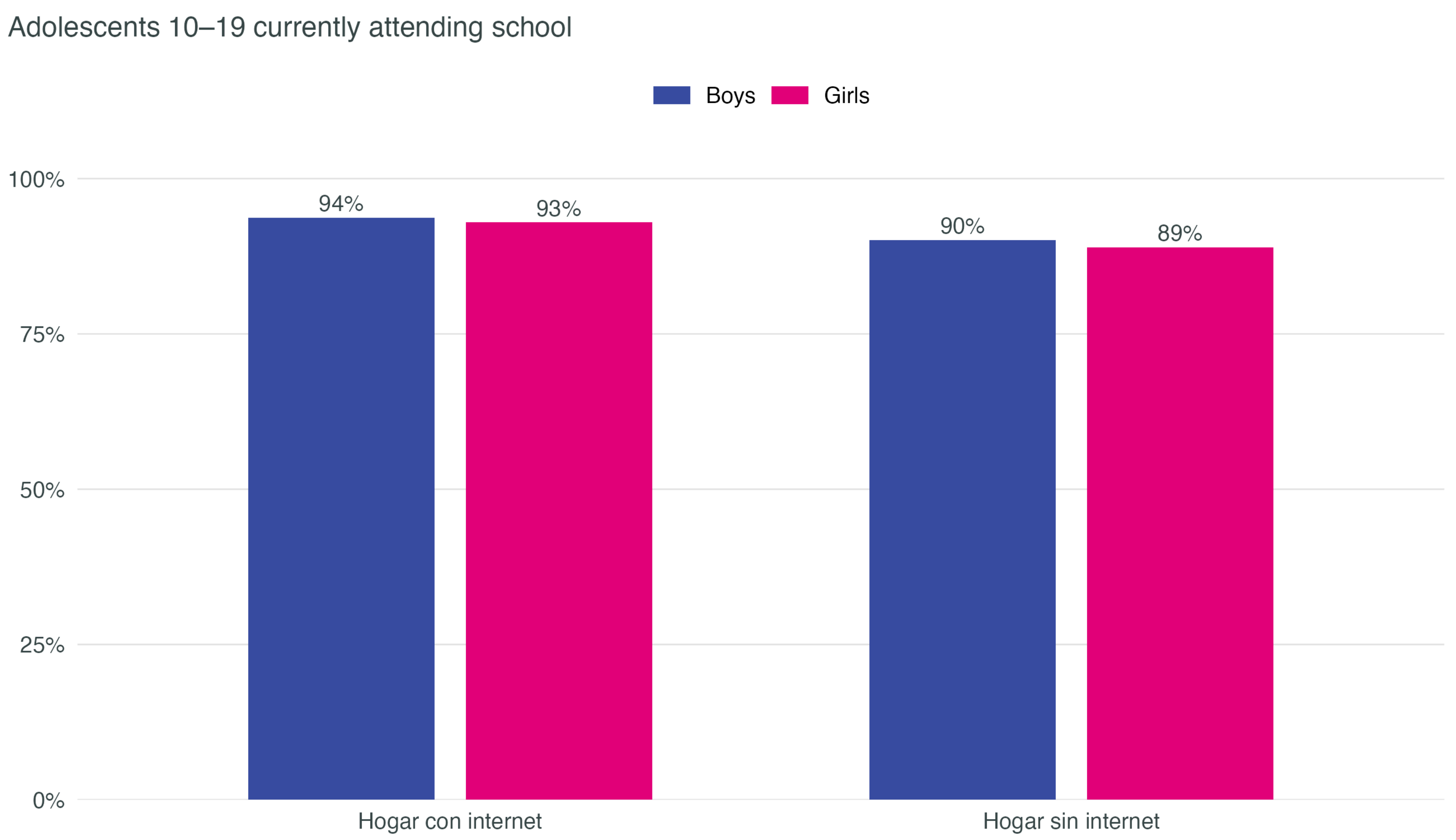
Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19.

Household digital access — no differences by sex



Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19.

Connected households show marginally higher school attendance for girls



Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–19.

2 · Evidence → Theory of Change

What the international evidence tells us

Consistent findings

- Isolated hardware rarely improves learning
  (De Melo et al., 2014; Cristia et al., 2017)
- Digital tools work **with structured pedagogy**
  (Gómez-Ruiz & Franco, 2023)
- Teacher training is the central mechanism
  (Olugbade et al., 2025; Bassi et al., 2016)
- Gender-responsive content improves **STEM aspirations and persistence**
  (Outes-León et al., 2020; Porter & Trzesniewski, 2020)

Parenting layer

- Parenting interventions reduce **physical and emotional violence against girls** (IRR 0.55)
  (Cluver et al., 2017 — South Africa)
- Digital parenting apps improve **non-violent practices and parent-daughter communication**
  (Janowski et al., 2025 — Tanzania)
- Restrictive gender norms at home are a structural barrier to learning, attendance and the STEM transition

Programme components — inputs and activities

Inputs

- Gender-responsive materials and safe access: digital content with female representation and free of gender bias
- STEM modules designed to retain and engage girls — a key gap in the STEM transition in Bolivia
- Connectivity and devices prioritised by need/vulnerability (rural, indigenous) rather than blanket delivery to schools

Activities

- Gender-transformative pedagogy and teacher training, with curricular integration of female role models
- Parental engagement for norm shift and linkage to STEM
- Girls’ empowerment, leadership and agency — beyond mere participation
- Socio-emotional learning and life skills, based on the literature review
- Mentoring and accompaniment

Programme components — outputs and outcomes

Outputs

- Data systems tracking sex-disaggregated participation and deeper learning outcomes
- Teachers trained, classrooms equipped, and content in use
- Girls integrated in STEM activities, with priority groups explicitly identified

Short-term outcomes

- Teacher practices that promote girls' active participation
- Support for STEM transition pathways
- Strengthened digital confidence and skills
- Safe and supportive learning environments

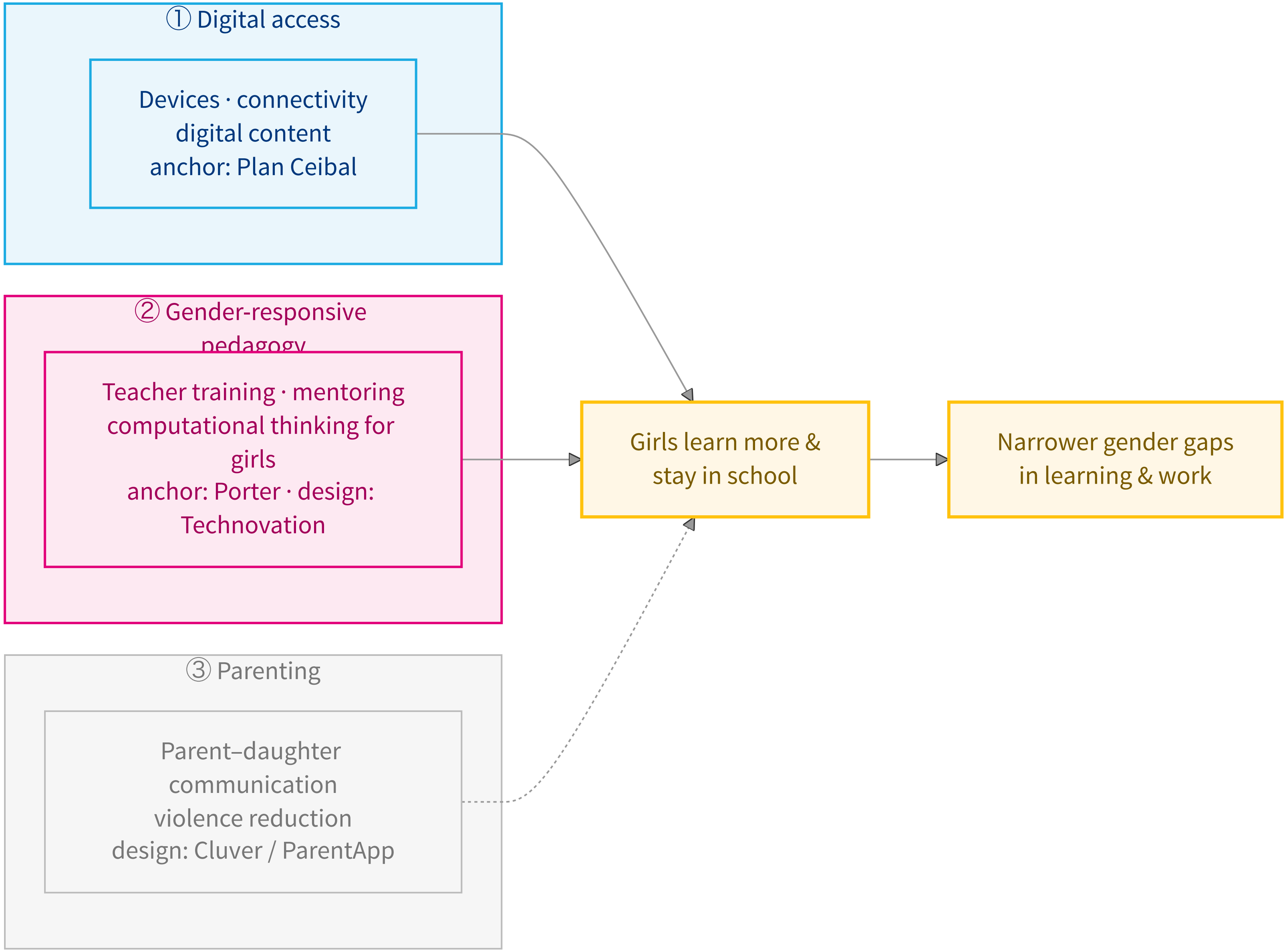
Medium-term outcomes

- Better attendance, progression and learning
- Higher interest and persistence in STEM
- Reduced household violence against girls, through the parenting layer
- Shifts in gender norms at home and school

Long-term outcomes

- Higher transition to tertiary education, particularly in STEM
- Better employability and reduction in wage gaps

What we are building — one programme, three layers



The quantitative anchor is **Porter** (South Africa) — a digital, secondary-age, majority-girls RCT — corroborated in-region by the **Mexico Learning Passport** (d=0.26). **Technovation Girls** (Chile) is the design template. The parenting layer (dashed) is **activated only when Bolivian data exists**.

3 · Projections — Aula Conectada in Bolivia

Direct anchors — digital education and gender norms

Programmes with measured effects **on the educational outcomes** Aula Conectada seeks to improve:

Programme	Country	External effect (95% CI)	ϕ_pop	ϕ_geo	ϕ_econ	ϕ_deliv.	n par.	Adjusted effect (95% CI)	Sig.
Mentalidad de crecimiento (Porter et al.)	Sudáfrica	-0.040 SD [-0.236, 0.156]	0.46	0.50	0.54	0.70	4	-0.003 SD [-0.020, 0.013]	—
ELA (Bandiera et al., 12-17 cohort)	Sierra Leona	7.2 pts [2.0, 12.5]	0.41	0.50	1.00	0.70	5	1.0 pts [0.3, 1.8]	✓
Plan Ceibal + programación (Gómez-Ruiz)	Uruguay	0.130 SD [0.032, 0.228]	0.49	0.85	0.30	1.00	4	0.016 SD [0.004, 0.028]	✓

Reading the CI: if zero is inside the interval, the anchor's original effect is not statistically distinguishable from zero — this reflects the source study's uncertainty, not an adjustment error. The **Sig.** column marks with ✓ the anchors that are statistically distinguishable from zero.

Indirect anchors — parenting layer (via SEM)

Parenting programmes that **do not act directly** on enrolment/attendance, but on the home environment (norms, violence). They enter the SEM as parameters of the parenting → home environment → educational outcomes pathway.

Programme	Country	Original outcome	ϕ_pop	ϕ_geo	ϕ_econ	ϕ_deliv.	Model use
ParentApp Tanzania (Janowski 2025)	Tanzania	see A9 (SEM)	0.16	0.50	1.00	0.70	input parameter for parenting → home environment
Parenting for Lifelong Health (Cluver et al. 2017)	Sudáfrica	see A9 (SEM)	0.97	0.50	0.54	0.70	input parameter for parenting → home environment

Cluver et al. (2017) reports effects on physical and emotional violence (IRR 0.55, not directly translatable to enrolment). Janowski et al. (2025) reports app engagement (modules completed). Effects propagate to educational outcomes via the SEM, not directly.

Key assumptions

The five ϕ factors are claims, not estimates

- $\phi_{\text{pop}} = 1$ when the source and target are identical in age and % rural
- $\phi_{\text{baseline}} = 1$ when Bolivia has full headroom; less when the indicator is already close to its ceiling
- ϕ_{geo} is banded by Haversine distance (1.00/0.85/0.65/0.50)
- ϕ_{econ} is capped at [0.30, 1.00] for stability
- ϕ_{delivery} reflects government implementation versus NGO/research

Robustness

- Multiplicative form is **conservative**: each $\phi \leq 1$, so the projection only attenuates the external effect, never amplifies
- Anchors are **only used** if their age range overlaps ≥ 25% with the target (10–19) and at least 2 demographic variables are comparable
- Sensitivity to alternative weightings (additive, distance-decay) in technical methods paper (in preparation)

Synthesis — expected returns from Aula Conectada

Learning effect
(priority group, ≈ 568K girls)

- **Anchor:** Porter (South Africa) — digital, secondary, 62% girls: **+0.27 SD** on mathematics
- **ϕ-adjusted to Bolivia: +0.16 SD** [0.00 – 0.27]
- **In-region corroboration:** Mexico Learning Passport RCT, **d = 0.26** (UNICEF, 2026)

Translated to lifetime earnings
(via Bolivia data)

- **+0.16 SD → +0.24 years** of schooling (Hanushek-Woessmann)
- **× 7.5%** Mincer return per year (EH 2024, **Bolivian women**)
- **→ ~$890** present value per girl (earnings channel only)

The earnings figure is a **floor**: agency, violence reduction and intergenerational gains are documented but not monetised. Honest downside: if the learning effect is null (contested literature), the return approaches zero.

4 · Costing

Unit costs — assumptions and sources

We construct **three scenarios (low / central / high)** from documented regional benchmarks. All values are annual per girl, in USD 2024.

Component	Low	Central	High	Source
Hardware amortised	30	48	60	Plan Ceibal Uruguay TCO: $192/4yr = $48/yr (ACM TechNews 2013); low = bulk procurement, high = harder-to-reach areas
School connectivity	12	20	30	Plan Ceibal connectivity+fibre: ~$20/child/yr; rural premium drives high scenario
Digital content & platform	5	10	15	IDB EdTech reviews 2018-2022: licensing + curated content cost $5-15/child/yr
Teacher training (annualised)	8	15	25	Bolivia secondary teacher salary $716/mo (boliviainformacion.com 2024); 20 days PD over 2 years amortised across ~1000 girls per cohort
Pedagogical support and M&E	5	8	12	Plan Ceibal admin: ~$27/yr (OLPC News 2010); fraction allocated to gender-pedagogy support
TOTAL per girl/year (USD)	60	101	142	Sum of components

Key notes: (1) Plan Ceibal reports total annualised cost of ~$75–100/child in implementation reports (OLPC News 2010, ACM 2013). (2) Bolivia teacher salaries 2024: ~Bs 3,206/mo primary (~$462) and Bs 4,947/mo secondary (~$716). (3) Rural permanence bonuses (Bs 659/mo) and per-rural-school bonuses (Bs 2,000) could reduce net cost in priority zones. (4) Figures exclude initial capital costs (electrical infrastructure, classrooms) — assumed pre-existing.

Programme scale — girls reached

Aula Conectada as investment in **adolescent girls 10–19**, with three priority-coverage scenarios:

Scenario	N girls	% of total	Coverage
Pilot (8 priority departments)	~80,000	8%	Rural+indigenous+poor
Intermediate scale	~280,000	29%	Broad priority subset
Full scale (all priority girls)	~568,000	58%	All girls with at least one vulnerability
Universal (all girls 10–19)	~977,000	100%	National coverage

Population figures derived from weighted EH 2024 (Table A1). The programme was designed for vulnerability-based targeting — we do not recommend universal coverage as it would dilute impact in households without digital access needs. **Central planning scenario** is full priority scale (~568,000 girls).

5 · ROI — Cost-benefit analysis

Monetised vs. broader returns

The BCR below monetises **only** the learning → earnings channel. It is deliberately conservative and should be read as the **floor**, not the ceiling, of expected returns.

What the BCR monetises

→ Learning gains (Porter, ϕ-adjusted)

→ Bolivia Mincer return (7.5%/year)

→ Discounted lifetime labour earnings (35 years)

What the BCR does NOT monetise *(documented, not summed)*

→ **Gender norms and agency**

→ **Household violence reduction** (Cluver PLH: IRR 0.55)

→ **Parental communication** (Janowski ParentApp)

→ **Intergenerational effects** on daughters and sisters

These channels are **documented but deliberately excluded** from the BCR — they would raise the return, not lower it, so the monetised figure is a floor.

Cost-effectiveness — one anchor, one effect

Anchored on a **single, well-matched programme** (Porter, South Africa): digital, secondary, 62% girls. No effects summed.

Metric (girls 10–19)	Central	Range
Learning effect (ϕ-adjusted)	+0.16 SD	0.00 – 0.27
Cost per girl (4 years)	$465	$360 – $850
Cost per +0.01 SD learning	**$29**	$13 – $53
Cost per +1 year of schooling	**$1,900**	—

In-region corroboration: **Mexico Learning Passport** RCT found **d = 0.26** on learning (UNICEF, 2026) — near Porter's 0.27. **Technovation Girls (Chile)** is the design template (no usable effect size).

*Low bound = **0.00 SD**: the strongest growth-mindset studies find near-null effects, so the floor reaches zero. "Central" = median cost and ϕ-adjustment; see A0.*

Benefit value chain — girls 10–19 in Bolivia

How the learning effect becomes a present-value benefit per girl:

Step	Calculation	Result
1 · Learning effect	Porter (S. Africa), ϕ-adjusted to Bolivia	**+0.16 SD**
2 · Years of schooling	× 1.5 yr/SD (Hanushek-Woessmann)	**+0.24 yr**
3 · Mincer return	× 7.5%/yr (EH 2024, Bolivian women)	**+1.8%** income
4 · Lifetime value	PV over 35 working years, disc. 3%	**~$890**
= Benefit vs. cost	$890 benefit ÷ $465 cost	**BCR ≈ 1.9**

Central scenario. If the effect is null (contested literature), **BCR ≈ 0** — the honest downside. Counts **only earnings**; agency, violence reduction and intergenerational gains documented but not added.

Key takeaways

1 · The gap is territorial, not enrolment. Enrolment is ~95% everywhere; the real gaps for girls are in **attendance, dropout, and digital access** — concentrated in rural, indigenous, and poor households.

2 · A proven model, adapted to Bolivia. Aula Conectada follows a digital, classroom-delivered design anchored on a single well-matched study (Porter, South Africa), with Technovation Girls (Chile) as the design template — no effects summed across programmes.

3 · Cost-effective on learning. ~$29 per girl for each +0.01 SD of learning; central **BCR ≈ 1.9** through the earnings channel alone. The honest downside: if the effect is null (contested literature), the BCR approaches 0.

4 · The monetised return is a floor. The central BCR (≈ 1.9) counts only earnings. Agency, violence reduction, and intergenerational gains are documented but **not** added — the true return is higher.

6 · Methodological Appendix

A0 · Cost-benefit analysis assumptions

- **Estimated costing.** Unit costs are projected from regional benchmarks (Plan Ceibal Uruguay, IDB EdTech reviews, Bolivia teacher salary scale). An official programme costing may refine the range — we recommend reviewing before scale-up decisions
- **Management and coordination costs.** Figures include administrative overhead (10-20%) but exclude in-country monitoring, inter-institutional coordination, and political/operational costs in remote areas. These could add 10-20% extra in rural scenarios
- **Benefits only via the education-labour channel.** We do not quantify benefits in agency, reproductive health, violence reduction, or domestic productivity — all potentially significant but harder to monetise defensibly
- **Mincer estimated from Bolivian data.** The return per year of schooling is estimated from EH 2024 (7.5%/year for women, 95% CI [6.6%, 8.4%]). The estimate does not incorporate the effect of schooling on the *probability* of employment — the total benefit is likely larger than projected here
- **3% discount rate** is the World Bank Education Sector standard; high sensitivity to discount rate (at 5% central BCR drops proportionally)
- **Analysis excludes positive externalities** on younger sisters, community norms, and intergenerational effects on future daughters — all documented in the literature but outside the scope of this conservative BCR

A1 · Data and sample

Source: Household Survey 2024 (EH 2024) — INE Bolivia · Collection November–December 2024 · National coverage, urban and rural · Unit of analysis: members of households in private dwellings.

Sample: 12,718 households · 39,497 persons · Analytical subsample: **6,254 adolescents 10–19** (≈ 977K girls and 1.05M boys expanded) · Files: Persona, Vivienda, Equipamiento.

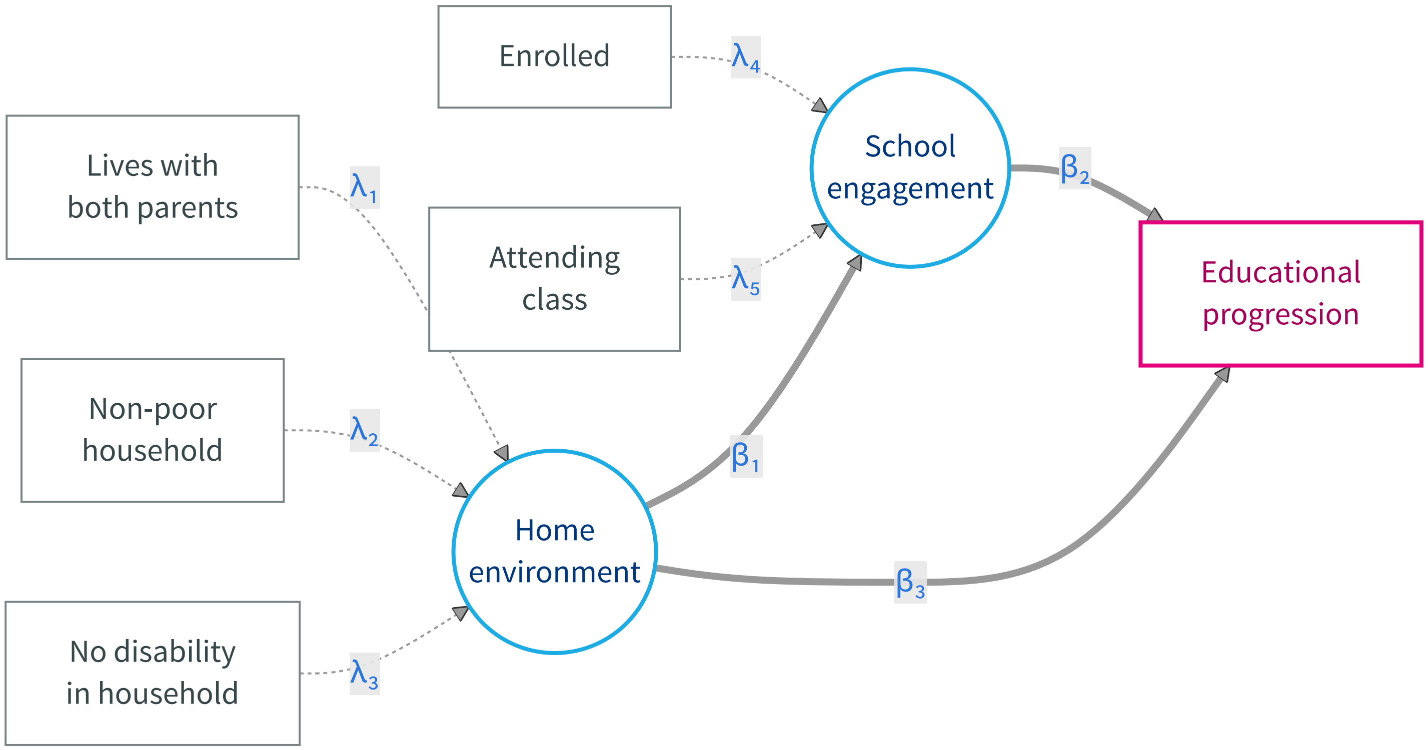
Survey design: Primary sampling units (`upm`), strata (`estrato`), expansion factors (`factor`). Estimates produced with `srvyr` (PSU `lonely.psu="adjust"`).

Key variables:

- **Sex** = `s01a_02` (1=male, 2=female) → girl = (sex == 2)
- **Age** = `s01a_03` (completed years)
- **Indigenous proxy** = childhood language OR spoken language in codes 1–37, 70, 71 (Bolivian indigenous languages)
- **Poverty** = INE-derived (`poor == 1`)
- **Disability** = Washington Group, severity ≥ 3 in any domain (seeing, hearing, walking, remembering)
- **HH internet** = `s06a_19` (Housing module) · **Digital equipment** (computer, smartphone, tablet) from Equipment module

Estimation: Weighted OLS using expansion factor `factor`; standard errors clustered at PSU level (`fixest::feols`).

A2 · Structural synthesis of the ToC — girls 10–19 (SEM)



Latent constructs (blue ovals) are measured by observed EH 2024 indicators (grey boxes); the final outcome is educational progression (pink). The parenting layer is **excluded** — Bolivia lacks contemporary parenting data (last full MICS: 2008) — and will be added when pilot or survey data exist. Equations and estimated coefficients in A11.

A3 · Empirical strategy

Base model (LPM with department fixed effects):

$$Y_{ihd} = \alpha + \beta_F F_i + \beta_R R_i + \beta_I I_i + \beta_D D_i + \beta_P P_i + \gamma X_{ihd} + \delta_d + \varepsilon_{ihd}$$

where F = Girl, R = Rural, I = Indigenous, D = Disability, P = Poor, X = controls (age).

Intersectional model with two-way interactions (× Girl):

$$Y_{ihd} = \alpha + \beta_F F_i + \gamma X_{ihd} + \sum_{k \in \{R,I,D,P\}} \theta_k \left(F_i \times k_i\right) + \delta_d + \varepsilon_{ihd}$$

Each θ_k measures how much larger or smaller the gender gap is in that subgroup. For example, θ_R captures whether the gender gap between rural girls and boys differs from the gap between urban girls and boys.

Intersectional model with three-way interactions (robustness):

$$Y_{ihd} = \alpha + \text{main effects} + \text{2-way} + \theta_{F \cdot R \cdot I} \left(F_i \times R_i \times I_i\right) + \delta_d + \varepsilon_{ihd}$$

and analogous for $F \times R \times P$ and $F \times I \times P$. Three-way interactions capture whether the gender gap in a compound intersection (e.g. rural indigenous girls) differs from what the two-way interactions would suggest separately.

Estimation by weighted OLS using `factor` (expansion factor); standard errors clustered at PSU level (`fixest::feols`).

A4 · Full coefficients table — main model

Variable	HH internet	Enrolled	Attending	Any HH device	Age-for-grade delay
Girl (1=yes)	0.013	-0.001	-0.005	0.004	-0.065***
	(0.011)	(0.008)	(0.008)	(0.005)	(0.014)
Rural	-0.241***	-0.068***	-0.070***	-0.044**	0.049**
	(0.039)	(0.020)	(0.020)	(0.022)	(0.021)
Poor (INE)	-0.142***	0.012	0.008	-0.016*	0.058***
	(0.022)	(0.008)	(0.009)	(0.008)	(0.016)
Indigenous	-0.007	-0.061***	-0.063***	-0.004	0.018
	(0.036)	(0.019)	(0.019)	(0.016)	(0.024)
Disability (WG3+)	-0.012	-0.472***	-0.469***	0.039***	0.356***
	(0.069)	(0.066)	(0.066)	(0.011)	(0.052)
N	7,517	7,517	7,517	7,517	7,517
R^2	0.150	0.141	0.140	0.034	0.058
Dept FE	Yes	Yes	Yes	Yes	Yes
Weights	Yes	Yes	Yes	Yes	Yes
Cluster SE (PSU)	Yes	Yes	Yes	Yes	Yes

Coefficients in decimal points (× 100 for pp). Robust SE in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

A5 · Full intersectional table

Interaction	Attending	Enrolled	HH internet	Any HH device
Girl × Rural	-0.047***	-0.043**	0.024	-0.008
	(0.018)	(0.018)	(0.030)	(0.013)
Girl × Indigenous	-0.056**	-0.052**	0.013	-0.006
	(0.022)	(0.023)	(0.037)	(0.013)
Girl × Disability	-0.032	-0.037	-0.050	-0.014
	(0.137)	(0.137)	(0.132)	(0.019)
Girl × Poor	-0.026	-0.020	0.025	-0.013
	(0.016)	(0.016)	(0.028)	(0.012)
N (each model)	7,517	7,517	7,517	7,517

Each row is the coefficient on the (Girl × subgroup) interaction in a separate regression. Coefficients in decimal points. Standard errors in parentheses. Robust SE clustered at PSU.

A6 · Heterogeneity analysis — regression forest

We complement the LPM with a regression forest (`grf::regression_forest`, Athey & Wager 2019) that estimates predicted outcomes conditional on the full set of demographic variables.

- **Training sample:** all adolescents 10–19 (not only girls) — lets the forest learn the gender effect endogenously
- **Forest:** `grf::regression_forest` with `honesty = TRUE` and 5,000 trees
- **Output:** variable importance and predicted subgroup gaps

The forest is **diagnostic, not causal** — there is no treatment in EH 2024 to estimate CATEs. Its value: identifying the demographic profiles where the (predicted) gap is largest, to guide programme targeting.

A7 · Formal foundation of the transportability approach

We assume that the mismatch between the source population S (the anchor) and the target T (Bolivia) decomposes into K observable dimensions, with multiplicative attenuation:

$$\hat{\beta_T} \approx \hat{\beta_S} \cdot \prod_{k=1}^{K} \phi_k(S, T), \qquad \phi_k \in [0, 1]$$

Glossary of ϕ_k factors:

- ϕ_{pop} — standardised age and % rural
- ϕ_{baseline} — outcome headroom
- ϕ_{geo} — Haversine distance from La Paz
- ϕ_{econ} — GDP per capita ratio
- ϕ_{delivery} — 1.0 (government) or 0.7 (NGO/research)

Assumptions:

- (A1) **Sufficient observables**: the 5 dimensions capture the relevant effect modifiers
- (A2) **Separability**: effects are approximately multiplicative
- (A3) **Monotonicity**: mismatch only attenuates, never amplifies
- (A4) **SE independence**: anchor variance propagates without perturbing the ϕ_k

A8 · Bolivia-adjusted projection methodology

Algorithm for each anchor:

1. Compute ϕ_{pop} as $1 - \sqrt{d_{\text{age}}^2, d_{\text{rural}}^2}$, where d are standardised differences between source and Bolivia
2. Compute $\phi_{\text{baseline}} = (Y_{\max} - Y_{\text{Bol}})/(Y_{\max} - Y_{\text{anchor}})$ when measurable (otherwise 1)
3. Compute ϕ_{geo} in Haversine bands from La Paz: ≤1,000 km → 1.00; 1,000–5,000 km → 0.85; 5,000–10,000 km → 0.65; >10,000 km → 0.50
4. Compute ϕ_{econ} as the Bolivia/anchor GDP per capita ratio, capped at [0.30, 1.00]
5. Compute ϕ_{delivery}: 1.0 if government implementer; 0.7 if NGO/research (Vivalt 2020)
6. Multiply: $\hat{\beta_T} = \hat{\beta_S} \cdot \prod \phi_k$
7. Propagate SE: $\text{SE}(\hat{\beta_T}) = \text{SE}(\hat{\beta_S}) \cdot \prod \phi_k$
8. 95% CI: $\hat{\beta_T} \pm 1.96 \cdot \text{SE}(\hat{\beta_T})$

Exclusion rules — age overlap <25% with 10–19, or <2 demographic variables comparable → anchor not projected.

Uncertainty — 95% CI propagated from the anchor SE. No synthetic noise on ϕ factors.

A9 · Consolidated evidence — anchors in use

Programme	Country	Age	Effect	Implementer	ToC arrow
Mentalidad de crecimiento (Outes & Sanc…	Perú	12-14	0.054 (0.030) SD	Government	Digital education
Mentalidad de crecimiento (Outes & Sanc…	Perú	12-14	0.135 (0.051) SD	Government	Digital education
Mentalidad de crecimiento (Porter et al.)	Sudáfrica	13-16	-0.040 (0.100) SD	NGO/Research	Digital education
ELA (Bandiera et al., 12-17 cohort)	Sierra Leona	12-17	7.250 (2.700) score_pts	NGO/Research	Gender norms
Irûmi — pensamiento computacional (Näsl…	Paraguay	7-8	0.089 (0.052) SD	Government	Digital education
Plan Ceibal SOLO HARDWARE (De Melo)	Uruguay	8-12	0.000 (0.050) SD	Government	Digital education
Plan Ceibal + programación (Gómez-Ruiz)	Uruguay	14-17	0.130 (0.050) SD	Government	Digital education
ProJoven (Ñopo et al.)	Perú	16-25	0.152 (0.060) ppts	Government	Labour market
ParentApp Tanzania (Janowski 2025)	Tanzania	0-17	0.130 (0.060) SD	NGO/Research	Parenting
Parenting for Lifelong Health (Cluver e…	Sudáfrica	10-17	0.100 (0.050) ppts	NGO/Research	Parenting

Complete list of external anchors used in the Bolivia projection (see `05_projections.R` and `R/anchors_demographics.R`). The full systematic review of 52 papers is in `data/lit_review_papers.csv`. The **ToC arrow** column indicates which arc of the Theory of Change each anchor feeds.

A10 · Robustness — three-way interactions (OLS)

To verify the heterogeneity detected by the regression forest, we estimate three-way interactions with OLS. If the forest correctly detects compound subgroups, the three-way coefficients should be significant for the outcomes with greatest heterogeneity.

Three-way interaction	Attending school	Enrolled	HH internet
Girl × Rural × Indigenous	-0.061 (0.045)	-0.052 (0.046)	0.050 (0.073)
Girl × Rural × Poor	-0.003 (0.036)	-0.008 (0.038)	-0.096 (0.069)
Girl × Indigenous × Poor	-0.002 (0.050)	0.005 (0.051)	0.096 (0.077)

How to read: each cell shows the three-way interaction coefficient and its standard error in parentheses. Significance: * $p<0.10$, ** $p<0.05$, *** $p<0.01$. Where no asterisks appear, the coefficient is not statistically distinguishable from zero at conventional levels. **Interpretation:** in this sample, the three-way interactions are not significant, indicating that the compound effect of belonging to multiple vulnerable groups (e.g. rural indigenous girl) is not substantially larger than the sum of effects of each group separately. Gaps accumulate roughly additively. This suggests well-targeted interventions on any axis (rural, indigenous, poverty) will reach girls who fit those characteristics.

A11 · SEM — equations and calibration

Approach: we estimate the ToC arrows for which EH 2024 has observed indicators (attendance, dropout, age-for-grade delay, household composition as a HomeEnv proxy). The **parenting-layer parameters** (β_3, β_5) are calibrated as **informative Bayesian priors** from the ϕ-adjusted anchors (Cluver, Janowski), not estimated from EH 2024. The structural diagram is in **A2**.

Pre-registered structural model

$$\text{GirlSkills} = \beta_2 \text{ClassPractice} + \beta_3 \text{HomeEnv}$$

$$\text{Attendance} = \beta_4 \text{GirlSkills} + \beta_5 \text{HomeEnv}$$

$$\text{Progression} = \beta_6 \text{Attendance} + \beta_7 \text{GirlSkills}$$

Parameter calibration

Parameter	Source
β_2	Outes-Sanchez + Gómez-Ruiz (ϕ-adjusted)
β_3, β_5	ParentApp (Janowski) + Cluver — **prior**
β_4	ELA (Bandiera) (ϕ-adjusted)
β_6	EH 2024 + pilot

Estimation via `blavaan` (informative Bayesian priors), latent scales fixed at $\lambda_1 = 1$ per construct, latent errors independent. Prior uncertainty propagates to the final BCR.

A12 · Bolivia Mincer estimation

Spec	Term	Coef	SE
A	aestudio	0.0559***	(0.0034)
A	exp_yrs	0.0376***	(0.0031)
A	exp_sq	-0.0007***	(0.0001)
A	female	-0.3257***	(0.0177)
A	rural	-0.4146***	(0.0484)
A	indigenous	-0.0101	(0.0227)
C	aestudio	0.0747***	(0.0046)
C	exp_yrs	0.0325***	(0.0041)
C	exp_sq	-0.0004***	(0.0001)
C	rural	-0.3814***	(0.0700)
C	indigenous	-0.0218	(0.0313)

Headline (Spec C, female-only): each additional year of schooling raises expected log monthly labour income by **7.5% [95% CI: 6.6% – 8.4%]**.

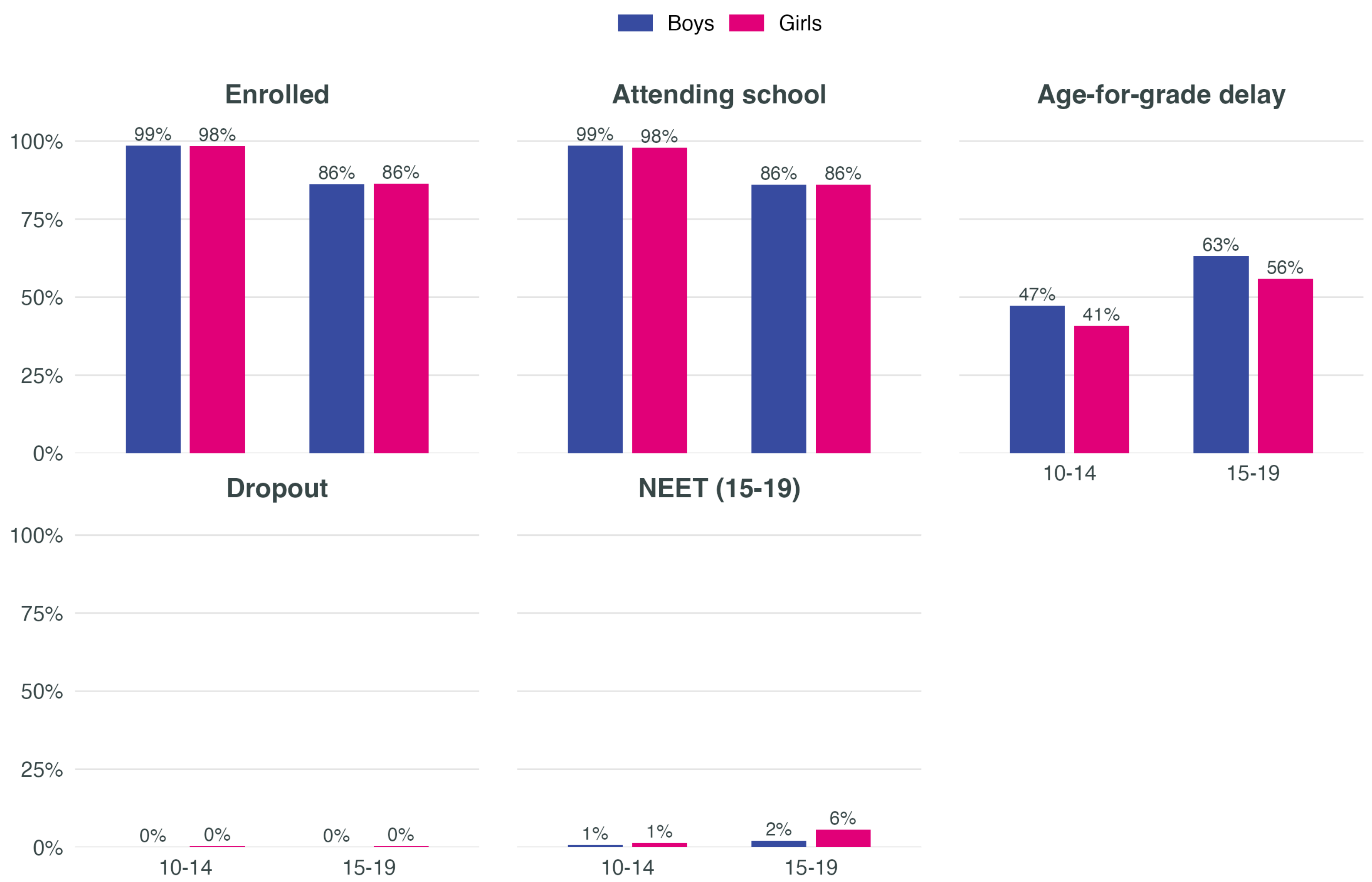
This figure enters the BCR directly as the Mincer return.

Sample: 13,009 adults 25-60 with positive labour income (5,327 women · 2,488 rural). Weighted OLS, PSU-clustered SE. Model: `log(ylab) ~ years_edu + exp + exp² + female + rural + indigenous`.

Caveats: (i) OLS on the employed sample suffers from selection bias — the return may be over-estimated for that sub-sample. (ii) We do not capture the effect of schooling on the *probability* of employment — this under-estimates the total benefit.

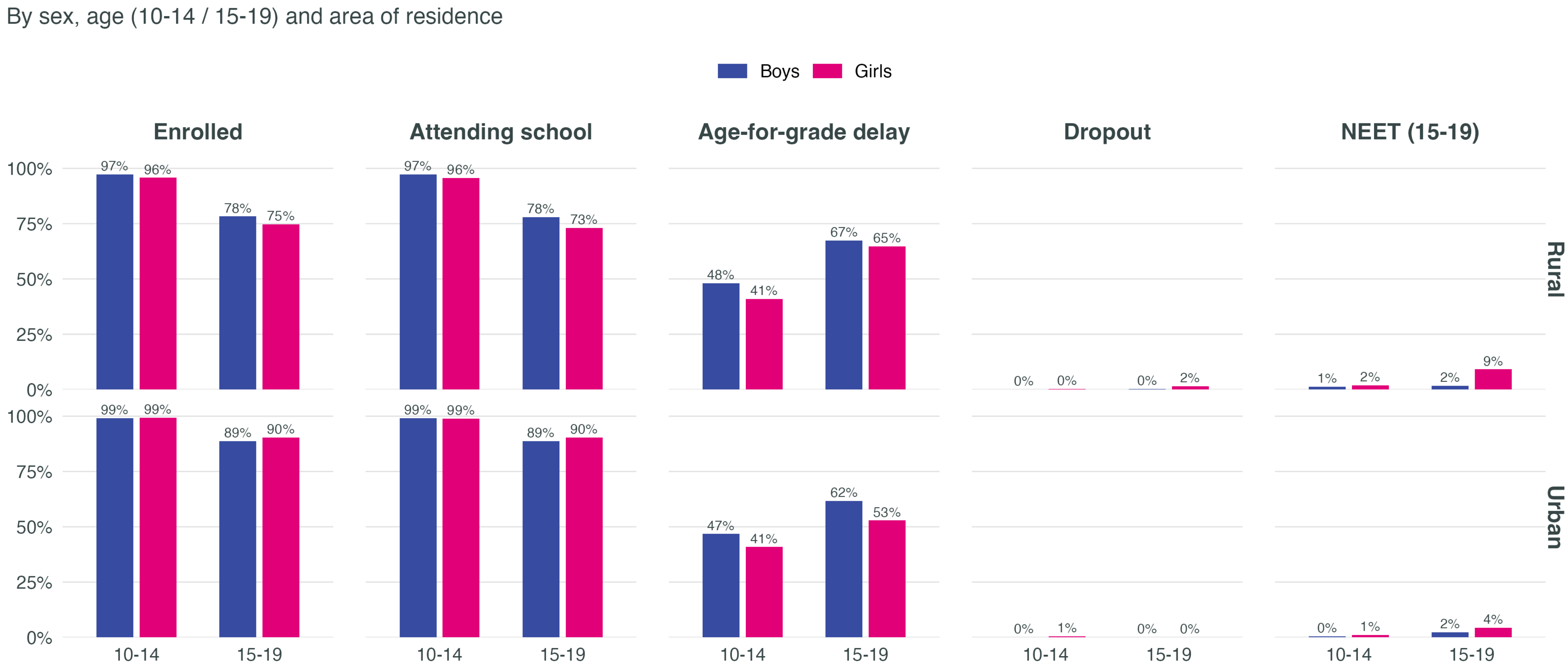
A13 · Education outcomes by age band — overall

Early (10-14) vs late (15-19) adolescence



Source: INE Bolivia – Household Survey 2024. Weighted estimates.

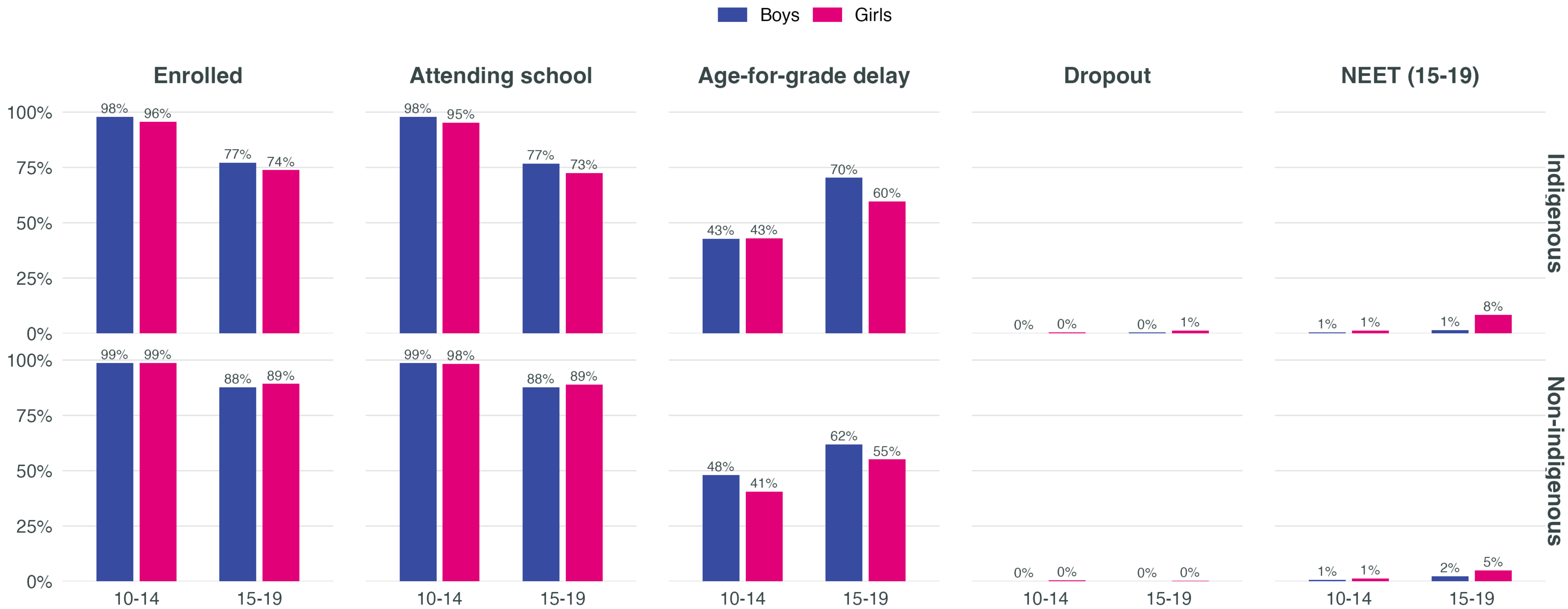
A14 · Education outcomes by age band — area of residence



Source: INE Bolivia – Household Survey 2024. Weighted estimates.

A15 · Education outcomes by age band — indigenous status

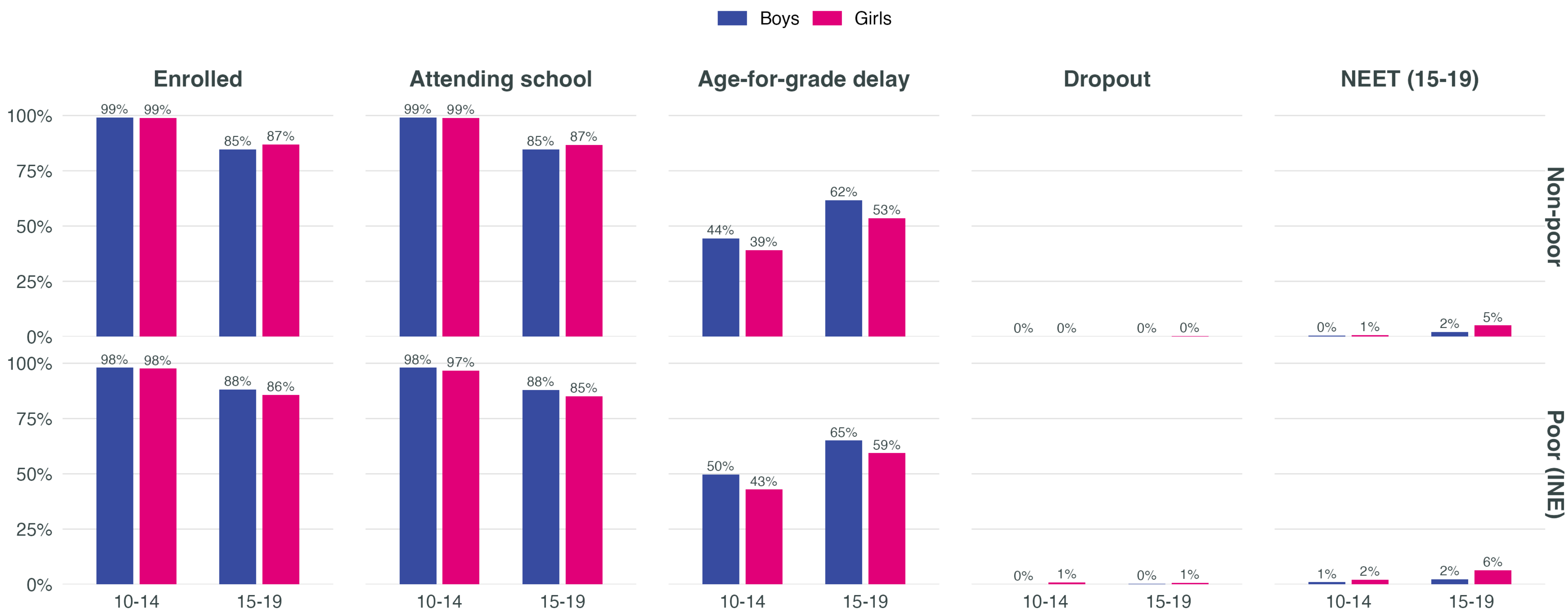
By sex, age (10-14 / 15-19) and indigenous status



Source: INE Bolivia – Household Survey 2024. Weighted estimates.

A16 · Education outcomes by age band — poverty status

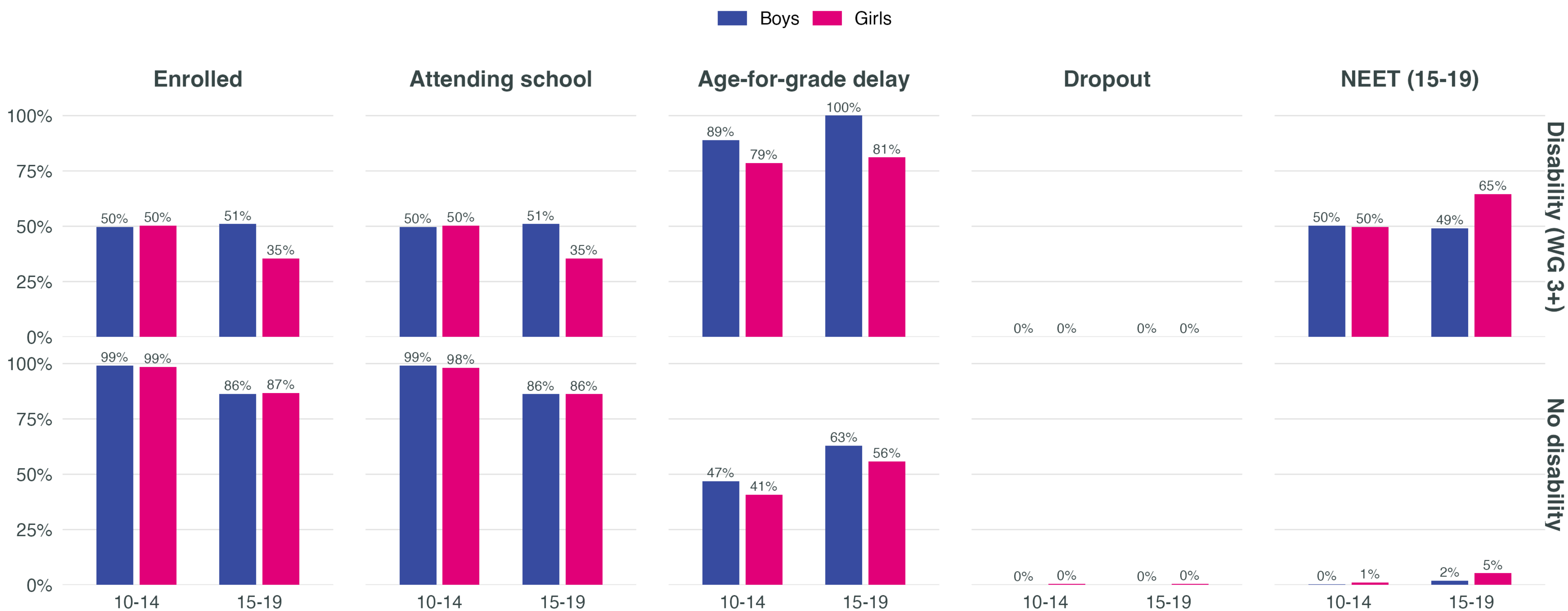
By sex, age (10-14 / 15-19) and poverty status



Source: INE Bolivia – Household Survey 2024. Weighted estimates.

A17 · Education outcomes by age band — disability

By sex, age (10-14 / 15-19) and disability



Source: INE Bolivia – Household Survey 2024. Weighted estimates.

A18 · Limitations

- EH measures **access to devices and internet**, not actual digital skills
- Household access **does not equal meaningful use** by the adolescent
- Disability sample size is **small** — interpret with caution
- **Cross-sectional** data: temporal trajectories cannot be observed
- Need for **direct measurement at the pilot level** to validate and refine projections

Thank you

Hernando Grueso, PhD

hgrueso@unicef.org